MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours) / MGU-BBA (Honours) / MGU-BCA (Honours)

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MG5DSCPSY300 – DYNAMICS OF RESEARCH IN PSYCHOLOGY

2024 Admission Onwards | Paper code: 26900447

ANSWER SCHEME

Part A – Multiple Choice Questions Answer all 1 mark each

1. Research methodology
2. Objectivity and honesty in reporting
3. Experimental research
4. Literature review → identify gap → establish need/significance → develop research problem
5. Objective analytical data
6. Research design
7. Confounded
8. Categorical and subjective measurement
9. It is a free and open-source statistical programming environment
10. Engages in the same activities as the people being observed

Part B – Short Essay Questions Answer any 6 Carries 5 marks each

11. A **sample** is a smaller representation selected from a population according to a plan. A sample-based measure is a **statistic**. **Sampling error** is the chance discrepancy between a sample statistic (e.g., sample mean) and the population parameter because only a subset is studied.

Reduction: (1) **Increase sample size** to improve precision and reduce variance. (2) Use **probability sampling**, such as simple random, stratified or systematic sampling: every member has a known, non-zero selection probability, limiting subjective selection and allowing sampling-error estimation. (3) Use **stratified random sampling**: divide heterogeneous populations into homogeneous strata (age, income, education), then sample randomly within strata to ensure subgroup representation. (4) **Optimize design and allocation** through appropriate proportional/disproportionate allocation and oversampling small or highly variable groups when required.

12. **Nominal:** Numbers/symbols label categories, without quantitative value, rank, meaningful origin or distance. Frequencies can be counted; arithmetic on labels is meaningless (e.g., averaging jersey numbers). Applications: demographic codes; jersey, roll and Aadhaar numbers; diagnostic categories/blood groups; consumer brand preferences.

Ordinal: Categories have a meaningful **rank order**, but intervals are unequal/unknown and absolute values absent. Rank 10 is not four times as good as rank 40; gaps between ranks 1–2 and 5–6 need not be equal. Applications: academic/percentile ranks; ordered agreement responses from Strongly Disagree to Strongly Agree; lower/middle/upper socioeconomic groups; mild/moderate/severe symptoms and disease stages.

13. **Probability and non-probability sampling**

Probability: Random selection gives every member a known, non-zero inclusion chance. It supports sampling-error estimation, inferential statistics, hypothesis testing and population estimation. Types: **simple random, stratified, cluster**. Example: randomly choosing 1,000 IDs from 10,000 university students gives each a 1-in-10 chance.

Non-probability: Selection relies on convenience, availability, judgement or criteria; probabilities are unknown. Selection bias limits statistical generalization, but it suits qualitative/exploratory work, low budgets and hidden populations. Types: **convenience, purposive, quota, snowball**. Example: interview available homeless youth at shelters and recruit peers through referrals.

14. A **variable** is a characteristic, condition or event that can be measured, manipulated or controlled and takes different values across people, situations or time (e.g., age, intelligence, anxiety, income). Unlike a constant, it varies; variables underpin hypotheses.

Independent variable (IV): Presumed cause/treatment manipulated, controlled or selected to examine an effect or relationship. **Dependent variable (DV):** Measured outcome or presumed effect. In a sleep-deprivation experiment, **4 versus 8 hours of sleep is the IV**; memory/logic test scores are the DV.

Extraneous variables: Other factors influencing the DV and threatening internal validity, such as prior intelligence, temperature or noise. **Control variables:** Factors held constant or otherwise controlled through design/statistical methods. Keeping noise and temperature constant prevents them distorting the sleep–performance relationship.

15. **Exploratory and descriptive research designs**

A **research design** is the conceptual structure within which research is conducted. **Exploratory design** addresses poorly defined or little-understood problems. As Kothari emphasizes, it discovers ideas/insights and clarifies problems or hypotheses. It is **flexible, open and discovery-oriented**; methods include literature review, experience surveys/expert interviews and insight-stimulating cases. It asks, "What is going on here?"

Descriptive design documents existing characteristics, functioning or prevalence without manipulating variables. It is **structured and pre-planned** to reduce bias and improve reliability, using surveys, standardized questionnaires, observation schedules and systematic sampling. It asks, "What are the characteristics or prevalence?" Exploratory research generates hypotheses; descriptive research measures existing conditions and establishes facts.

16. A **hypothesis** is a tentative, testable proposition predicting relationships between variables, formulated to examine logical/empirical consequences. It bridges **theory and empirical observation**, providing direction and specificity.

Functions: (1) Identifies variables and study scope. (2) Guides design, tools, psychological scales and relevant data collection. (3) Prevents aimless data collection and conserves resources. (4) Guides testing and interpretation through null/alternative and directional/non-directional hypotheses, using suitable t-tests, ANOVA or correlation. (5) Promotes **objectivity and replication** by specifying expected outcomes operationally before data collection. As Kothari emphasizes, hypotheses connect theory with systematic scientific inquiry.

17. **Scientific observation** systematically and purposefully records actual behaviour, interactions and events. It supplies first-hand evidence when self-reports are inaccurate/unavailable, disclosure is limited, or non-verbal/social behaviour matters. It describes phenomena, generates hypotheses, verifies other data and preserves context.

Types of observation: (1) Participant/non-participant: The researcher joins group activities to understand customs, relationships and behavior, or observes externally with less involvement and potentially greater objectivity. (2) Structured/unstructured: Predetermined categories or checklists improve reliability and comparability, while flexible recording supports exploration. (3) Controlled/naturalistic: Behavior is observed under arranged, standardized conditions or in real-life settings without interference. (4) Direct/indirect: Behavior is observed as it occurs or studied through traces, records and reports. (5) Disguised/undisguised: Participants are unaware or aware of observation. These methods provide rich, realistic, detailed data but require attention to observer bias, observation-induced behavioral changes and ethical concerns.

18. **Research misconduct**

Ethics requires **honesty, accuracy and respect for intellectual work**. Misconduct during proposing, conducting, reviewing or reporting research distorts knowledge and public trust.

Fabrication: Inventing data, observations, results or sources; e.g., reporting responses never collected. **Falsification:** Manipulating materials, procedures, equipment or data, including selective alteration/omission; e.g., deleting unfavourable responses. **Plagiarism:** Presenting others' words, ideas, data, methods or results without acknowledgement, including copying and uncited paraphrasing; citation failures may be unintentional. **Prevention:** Report honestly, preserve records, cite correctly and credit contributors/prior authors.

Part C – Essay Questions |Answer any 2 Carries 15 marks each

19. **Steps in the research process**

The **research process** is a systematic sequence of interrelated activities: identify a problem, collect and analyse evidence, interpret findings and report conclusions.

1. Define the research problem: Identify and clearly formulate an investigable problem. Earlier studies and experiments reveal unresolved findings, unexplored areas or promising results not followed up. Convert a broad interest into a precise problem.**2. Review the literature Conduct an extensive survey of relevant research:** Consult **abstracting/indexing journals, published/unpublished bibliographies, academic journals, conference proceedings, government reports and books**. Carefully examine related investigations; use libraries and follow references from one source to another to establish background and gaps.**3. Formulate working hypotheses**: State **clear, tentative propositions** whose logical/empirical consequences can be tested. Hypotheses provide the focal point, guide evidence requirements and influence analytical and the relevance/quality of data.**4. Prepare the research design:** Develop the **conceptual structure and plan** for obtaining relevant evidence efficiently, maximizing information while minimizing effort, time and money. Purposes include **exploration, description, diagnosis and experimentation**.**5. Collect data:** When existing data are inadequate, collect appropriate primary data through **experiments or surveys**, considering resources, time and cost. Experiments obtain measurements to examine hypotheses. Survey methods include: **i. Observation:** Investigator records information directly without interviewing respondents. **ii. Personal interviews:** Obtain answers to predetermined questions through direct interaction and a planned procedure. **iii. Telephone interviews:** Contact respondents by telephone. **iv. Mailed questionnaires:** Respondents complete and return carefully prepared questionnaires. A **pilot study** identifies weaknesses and improves the instrument before full use. **v. Schedules:** Trained enumerators visit respondents and complete schedules from their replies.**6. Analyze data:** Organize raw data through **classification, coding and tabulation**. Condense unwieldy data into manageable, meaningful categories and tables, apply suitable statistical

procedures and draw inferences to examine hypotheses.**7. Interpret findings and write the report**: **Interpretation** explains findings in relation to hypotheses, earlier evidence and theory. Repeatedly supported hypotheses may contribute to generalization and theory-building. Without an initial hypothesis, findings can be explained using relevant theory. Interpretation may generate new questions and further research. Prepare a clear, carefully written account of procedures and conclusions. Report layout comprises **preliminary pages, main text and end matter**. The sequence links problem formulation and evidence collection with analysis, explanation and communication.

20. A **literature review** is a systematic survey and critical evaluation of existing research related to a problem. It contributes throughout research, beginning when a question is first considered. Establishes background/existing knowledge; reveals **gaps and contradictions**; clarifies problems, objectives and hypotheses; builds theoretical/conceptual frameworks; guides variables, design, sampling, tools and statistics; avoids unnecessary duplication; strengthens originality/credibility; enables comparison of new and previous findings. **Primary sources:** Original peer-reviewed research articles, theses/dissertations, conference papers, patents, official research reports and raw datasets provide first-hand evidence and methodological detail. **Secondary sources:** Textbooks, review articles, meta-analyses, handbooks, encyclopedias and edited books summarize or interpret original work. **Tertiary/locating resources:** Library catalogues, bibliographies, indexes, abstracts and databases/search tools: **Google Scholar, PubMed, Scopus, Web of Science, JSTOR and ERIC**. **Other sources:** Government/census publications, policy documents, annual reports, newspapers, professional association publications and grey literature, including working papers and conference proceedings.

Review methods: Narrative/traditional: Broad, flexible contextual overview, less structured. **Systematic:** Predefined protocol, explicit searches, inclusion/exclusion criteria, quality appraisal and synthesis reduce bias. **Scoping:** Maps the extent/nature of broad or emerging literature, concepts and gaps. **Meta-analysis:** Statistically combines comparable quantitative results into an overall estimate. **Meta-synthesis:** Integrates qualitative findings. **Theoretical:** Examines/compares theories. **Critical:** Evaluates strengths, weaknesses and assumptions. **Procedure:** Define topic; select keywords; search databases; screen relevance/quality; organize thematically or chronologically; compare studies, identify gaps and synthesize critically; cite accurately to avoid plagiarism.

21. Reliability and validity are the two essential qualities of any data-collection tool, such as a questionnaire, interview schedule, test, rating scale, checklist, or observation schedule. **Reliability** means the consistency, stability, and dependability of a tool; a reliable tool produces similar results when used repeatedly under similar conditions, with the same respondents, or by different trained observers. **Validity** means the accuracy or truthfulness of a tool; it measures whether the instrument actually measures the concept it claims to measure. Both are necessary because unreliable or invalid tools produce inaccurate data, leading to wrong analysis, interpretation, and conclusions. Reliability is important because it reduces random errors and increases confidence in the collected data. A reliable questionnaire or test gives nearly the same score when administered again after a short interval, when scored by different examiners, or when its items consistently measure the same characteristic. The major types are test–retest reliability (stability of scores over

time), inter-rater reliability (agreement between observers or scorers), internal consistency reliability (consistency among items of a scale), and parallel-form reliability (equivalence of two versions of the same tool). Reliability is commonly tested through the **test–retest method, split-half method, Cronbach's alpha, and correlation coefficients**. It is

especially important in **surveys, psychological testing, experiments, and observation** because it makes results replicable and strengthens the scientific value of the study. Validity is important because it ensures that the tool measures the intended variable rather than some unrelated characteristic. For example, a test designed to measure mathematical

ability is valid only if it genuinely measures mathematical ability and not merely language skill or memory. The main types are face validity (whether the tool appears appropriate), content validity (whether items cover all aspects of the concept), construct validity (whether the tool measures the underlying theoretical construct), and criterion validity

(whether the tool's results agree with an established measure or predict a future outcome). Validity can be established through **expert judgement, pilot testing, comparison with standardized instruments, correlation analysis, and factor analysis.** A tool may be reliable without being valid for example, a weighing machine may consistently show an incorrect weight but a valid tool must also be reliable. Thus, reliability and validity together ensure accuracy, consistency, credibility, and generalizability of research findings, making them indispensable in the construction and use of data-collection instruments.

22. Trustworthiness establishes rigour in qualitative studies of meanings, experiences and social contexts. **Credibility** is confidence in the truth of findings; **dependability** is the consistency, stability and traceability of the process and findings. They correspond broadly to quantitative validity and reliability, while recognizing contextual interpretation.

1.Triangulation: Cross-check findings using multiple data sources, methods, investigators or theories; e.g., combine **interviews, observations and documents**. **2. Prolonged engagement:** Spend sufficient time with participants to understand the setting, build rapport and reduce distortions. **3. Persistent observation:** Repeatedly examine the most relevant features to understand important patterns and contextual details. **4. Member checking:** Return

transcripts, themes, interpretations or conclusions to participants to confirm their accuracy and correspondence with experience. **5. Peer debriefing:** Discuss procedures and interpretations with colleagues to question assumptions and examine explanations. **6. Reflexivity:** Acknowledge the researcher's personal biases, assumptions and influence on data collection and interpretation. **7. Negative-case analysis:** Examine evidence contradicting emerging themes instead of ignoring it; refine interpretations accordingly. To establishing the dependability using the strategies: **1. Audit trail:** Keep detailed, reviewable records of **research design, sampling decisions, data-collection procedures, field notes, transcripts, coding decisions, changes in the research plan, analytical decisions and the path to conclusions**. Show transparently how data became findings.

2. Peer examination: Another researcher reviews procedures/analysis or codes the same data to assess consistency and the evidential basis of interpretations.

3. Code–recode procedures: Revisit and recode data to check coding consistency. Clear methods documentation and field notes make analysis traceable.

C. Related dimensions of trustworthiness

Confirmability: Findings must be grounded in participants' data rather than researcher preferences or bias. Reflexivity and transparent records demonstrate the evidential basis of conclusions.

Transferability: Provide **thick description** of participants, setting and context, enabling readers to judge applicability to other settings rather than assuming universal generalization.

Together, **triangulation, prolonged engagement, persistent observation, member checking, peer debriefing, reflexivity, negative-case analysis, audit trails and thick description** demonstrate accurate interpretation, systematic procedures and overall qualitative rigour.